

Tribhuvan University
Institute of Science and Technology
2074

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (MTH:112)
(New Courses)

Full Marks: 80
Pass Marks: 32
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all questions.

Group A (10×2=20)

1. (a) A function is defined by $f(x) = \begin{cases} x + 2 & \text{if } x < 0 \\ 1 - x & \text{if } x > 0 \end{cases}$, calculate $f(-1)$, $f(3)$, and sketch the graph. (5)
(b) Prove that the $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.
2. (a) Find the derivative of $f(x) = \sqrt{x}$ and to state the domain of f'
(b) Estimate the area between the curve $y^2 = x$ and the lines $x=0$ and $x=2$.
3. (a) Find the Maclaurin series for e^x and prove that it represents e^x for all x .
(b) Define initial value problem. Solve that initial value problem of $y' + 5y = 1$, $y(0) = 2$.
(c) Find the volume of a sphere of radius r .
4. (a) For what value of x does the series $\sum_{n=1}^{\infty} \frac{(x-3)^n}{x}$ converge?
(b) Calculate $\iint_R f(x, y) dA$ for $f(x, y) = 100 - 6x^2y$ and $R: 0 \leq x \leq 2, -1 \leq y \leq 1$.

Group B (5×4=20)

Attempt any ten question

5. If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{3-x}$, find $g \circ f$ and $f \circ g$.
6. Use continuity to evaluate the limit, $\lim_{x \rightarrow 4} \frac{5+\sqrt{x}}{\sqrt{5+x}}$.
7. Verify Mean value theorem of $f(x) = x^3 - 3x + 3$ for $[-1, 2]$.
8. Sketch the curve $y = x^3 + x$
9. Determine whether the integral $\int_1^{\infty} \frac{1}{x} dx$ is convergent or divergent.

Group C (5×8=40)

- 10.** Find the length of the area of the semicubical parabola $y^2 = x^2$ between the point (1,1) and (4,8).
- 11.** Find the solution of $y'' + 6y' + 9 = 0, y(0) = 2, y(0) = 1$.
- 12.** Test the convergence of the series $\sum_{n=1}^n \frac{n^n}{n!}$.
- 13.** Define cross product of two vectors .if $a=i+3j+4k$ and $b=2i+7j+5k$, find the vector $a \times b$ and $b \times a$.
14. Define limit of a function . find $\lim_{x \rightarrow \infty} (x - \sqrt{x})$.
15. Find the extreme value of $f(x, y) = y^2 - x^2$